Li Guangjie

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EDUCATION

Hong Kong University of Science and Technology 2026-2027

- Master of Science in Environmental Engineering and Management
- Research Interest: Building Energy Systems For Model Predictive Control supervisor: Wang Zhe

Guangzhou University 09/2022-07/2026

- Major: Building Environment and Energy Engineering
- Degree: Bachelor of Engineering
- Average Score: 86/100
- **2024-2025 Academic Year, National Scholarship--Academic Evaluation Ranking: 9/213**
- **2024-2025 Academic Year Major Ranking: 8/91, Class Ranking: 1/35**

Core Courses:

- *Renewable Energy and Low-Carbon Cities (95), Intelligent Buildings (95), Cooling and Heating Source Engineering (93), Heat and Mass Transfer Mechanism and Equipment (90), Heating Engineering (93), Fluid Piping (90), Building Electrical (96), BIM Techniques and Applications (92),Air Conditioning Engineering (86, highest in class), C Programming (95), Automatic Control (81), Building System Automation (80)*

AWARDS & HONORS

- **Academic Progress Award** for the 2024-2025 Academic Year
- **Second-Class Scholarship** for the 2024-2025 Academic Year
- **Third-Class Scholarship** for the 2025-2026 Academic Year
- **Outstanding Student Leader** for the 2024-2025 and 2025-2026 Academic Year
- **Third Prize** in the New Star Cup Structural Design Competition of Guangzhou University

INTERNSHIP EXPERIENCES

China Academy of Building Research- Institute of Environment and Energy 11/2025-05/2026

Assistant to Senior Engineer

- Assisted in developing a preliminary Large Language Model (LLM) diagnostic tool for building energy audits. Refined data request checklists and diagnostic workflows, significantly reducing manual auditing efforts and establishing a standardized diagnostic protocol.
- Served as a core member in conducting field tests for Midea & Aux Variable Refrigerant Flow (VRF) units in the extreme dry-hot region of Turpan. Processed and analyzed over 1 million raw sensor data points; authored data logging protocols, technical reports, and optimization analyses.
- Participated in energy audits for 8 commercial buildings (offices, malls, hotels). Cleaned and processed over 100,000 electricity consumption data points to support energy modeling and report generation.
- Conducted on-site environmental testing and material chamber experiments for the Pazhou Near-Zero Emission Building project, providing quantitative scoring support.
- Co-authored a dual-carbon industry report focusing on business model upgrades from project-based technical services to platform-based carbon asset operations.

Guangdong Tongfang Ruifeng Co., Ltd. 12/2024-01/2025

Intern Group Leader

- Led a team in the production of HVAC units for Tongchuan New Energy project; involved in entire process from component processing (fans, filters, evaporators) to final assembly.

- Engaged in and mastered the design principles and selection criteria for core components of the air conditioning unit (compressor, surface cooler, fan), such as the requirement to enhance dehumidification performance for IDC data centers.

Guangdong Xiangsheng Construction Engineering Co., Ltd. 08/2024

Construction Engineering Assistant

- Assisted in on-site construction management, installation of construction equipment, and performed stability checks on formwork supports and steel reinforcement binding; paid particular attention to the reinforcement configuration and inspection of critical structural components such as beam-column joints and cantilever sections.
- Conducted on-site verification against construction drawings, observed structural implementation, and communicated with installation personnel to ensure accurate execution in accordance with design specifications.
- Responsible for the configuration and inspection of electrical systems, including the pre-burial and surface mounting of wire conduits (PVC and JDG pipes) for lighting, sockets, and power circuits, as well as the installation of distribution boxes.

Wuhan Zhongke Kechuang Engineering Inspection Co., Ltd. 07/2024

Engineer Assistant

- Based on over 200 sets of foundation settlement data, developed a bearing capacity prediction model to support engineers in efficiently evaluating the quality of pile foundation construction.
- Performed non-destructive testing (penetrant and ultrasonic testing), operated static load testing equipment, strain acquisition systems, and ultrasonic flaw detectors, independently completing data collection for foundation settlement and weld defect detection in steel structures.

ACADEMIC PROJECTS

Experimental Study on the Energy Efficiency of Residential Multi-Split Air Conditioners under Different Temperature Setting Modes and Indoor Unit Operation Quantities 05/2026

- Submitted to Journal of Appliance Science & Technology, under review.
- Measured and calculated cooling EER of a 1-to-4 DC inverter VRF system under different temperature modes and indoor unit ON/OFF combinations.
- Analyzed influencing factors using Excel; provided practical recommendations for energy-efficient design and user behavior in residential VRF systems.

Differentiated Thermal Comfort Control Scheme 04/2025

- Provincial-level College Student Innovation & Entrepreneurship Program.
- Proposed time-based transition strategies and spatial zoning; introduced visual recognition and trajectory prediction for non-core occupants' thermal comfort while protecting privacy via anonymization.
- Applied differentiated control strategies for core vs. non-core personnel; integrated phased power regulation and renewable energy to reduce building energy consumption.
- Combined multiple sensors and AI algorithms to build an intelligent HVAC regulation system.

Energy Efficiency Inspiration from Traditional Architecture to Modern Architecture 07/2024

Insights from Ancient Architectural Structures on Energy Efficiency and Thermal Comfort in Modern Buildings

- Conducted on-site experimental data testing, including measuring and recording indoor temperature, humidity, and air velocity in ancient ancestral halls.
- Analyzed the thermal environment of the buildings based on the collected data, utilizing tools such as Excel and MATLAB for data interpretation and modeling.
- Reviewed academic literature and references related to human thermal comfort, analyzed the influence of various ancient architectural structures, and compared the energy-saving potential of traditional designs with modern building energy consumption, summarizing key findings accordingly.